

CS & IT ENGINEERING

Algorithm

Miscellaneous

Lecture No. 04



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Topic

OBST $\rightarrow$ DP
Backtracking

Topics to be Covered



Topic

Topic

Topic

Branch-Bound

Set

↳ Kruskal Algo

Tip & Tricks :-



About Aditya Jain sir



1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram



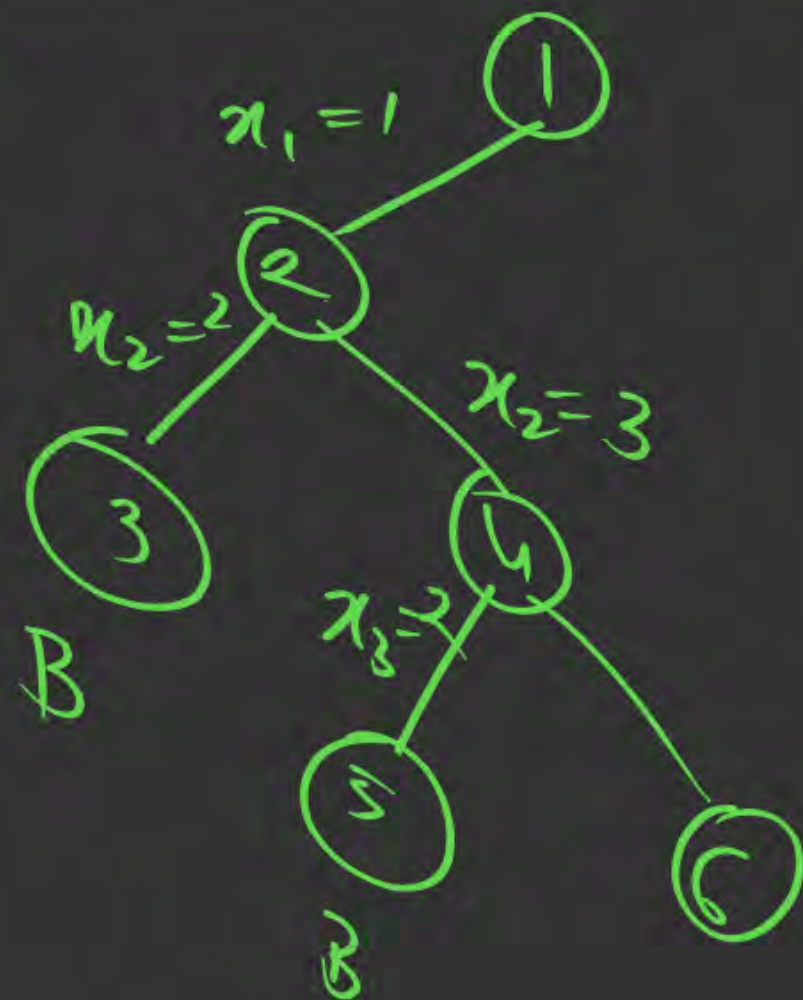
Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

Branch-Bound $\Rightarrow$ BFS + Bounding

Backtracking: DFS + Bounding func

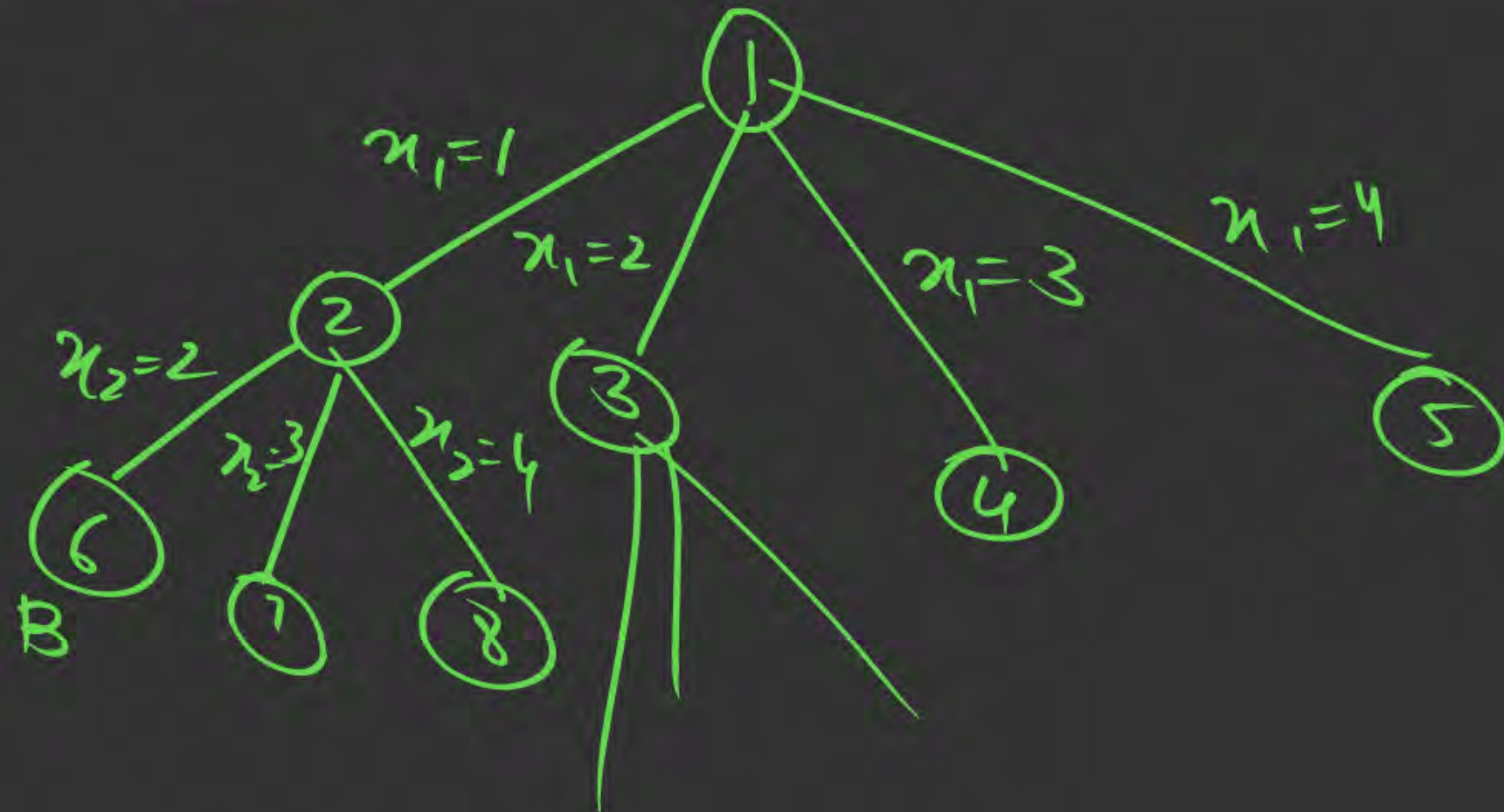
eg:- 4-Queens Prob

DFS



Branch-Bound: BFS

1	2	3	4	5	7	8
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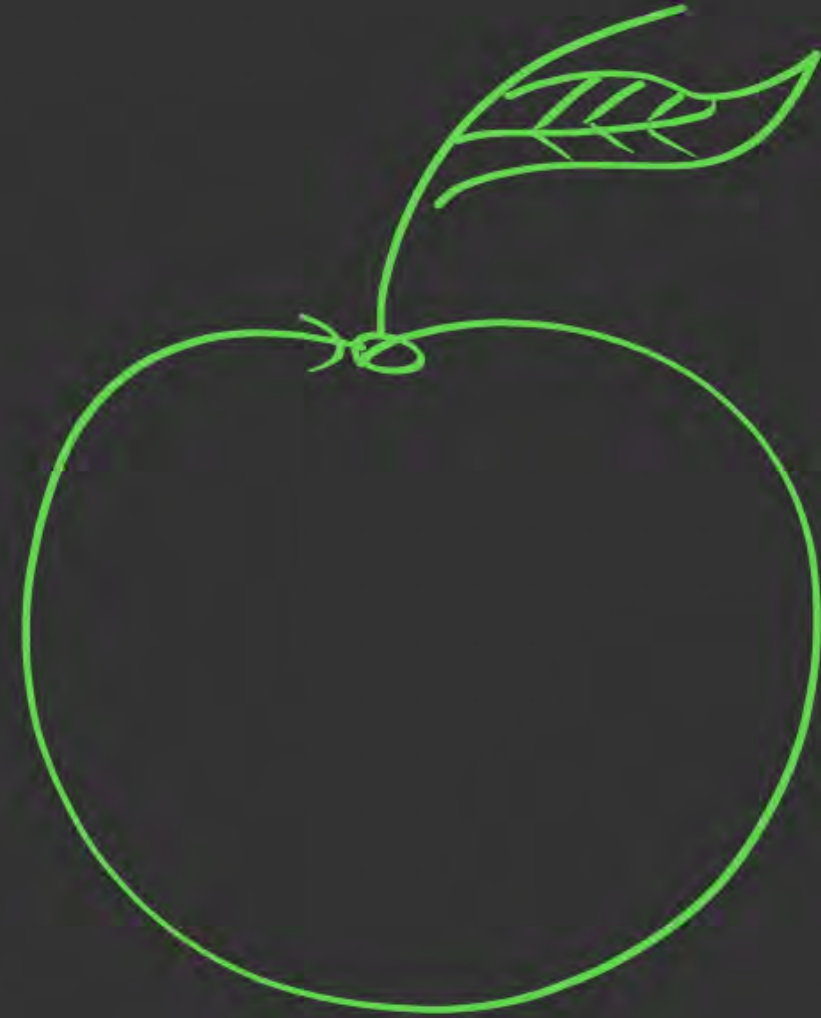


Backtracking

- 1) n -Queens
- 2) SOS
- 3) Graph Coloring
- 4) 0/1 Knapsack

Branch-Bound

- 1) n-Queens
- 2) 0/1 Knap
- 3) TSP
- 4) TSP





Topic: Sets



Unordered

~~Wonder~~ collection of non-repeated/ non-duplicate/ distinct elements.

Example:

$$S_1 = \{1, 5, 7, 2\}$$

There is no number including in sets.

No indexing





Topic: Sets



Venn diagram:

1. Disjoint sets: Mutually exclusive sets
~~number~~ common elements
no

Equation:

S1: set of even numbers = {2, 4, 6}

S2: set of ~~even~~ numbers = {1, 3, 5}

S1 $\cap$ S2 ^{odd} = ϕ (empty set)
~~~~~

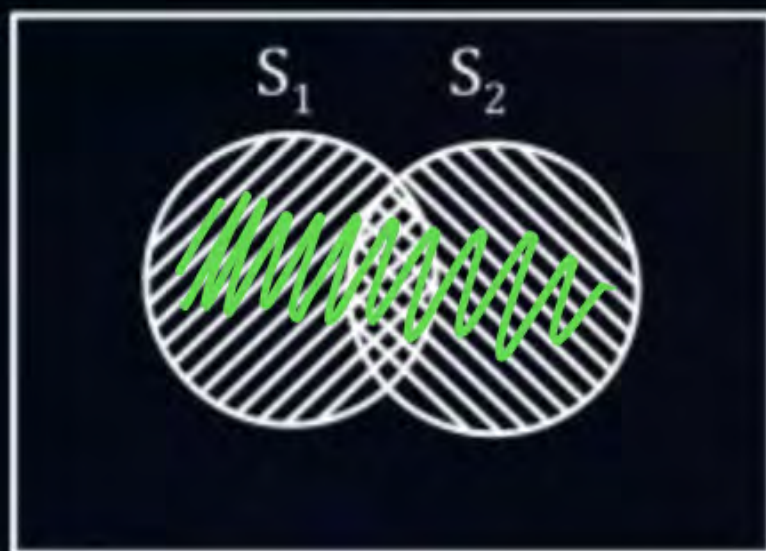


## Topic: Sets

### 1. Union:

$$S_1 \cup S_2 = \{1, 2, 3, 4, 5, 6\}$$

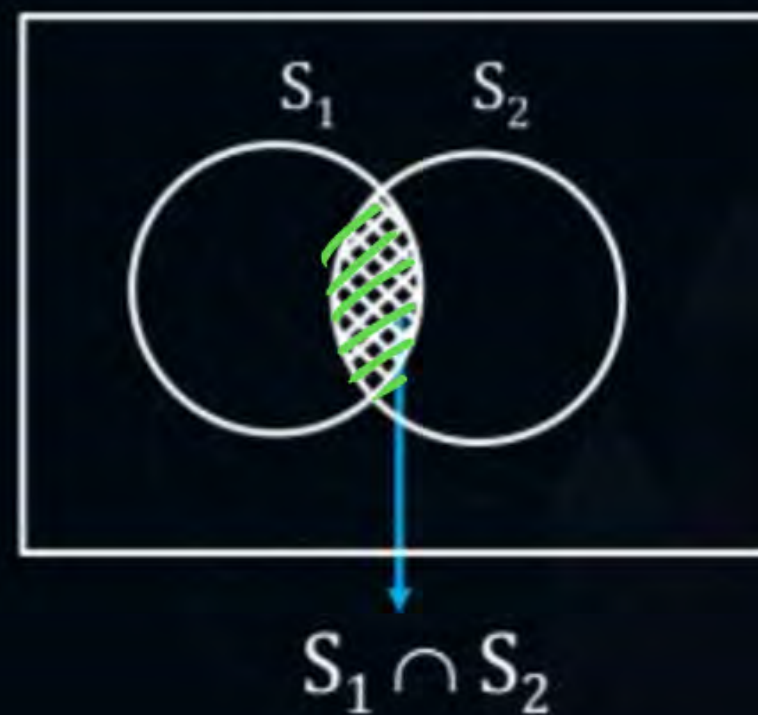
( $\cup$ )



### 2. Intersection

( $\cap$ )

$$S_1 \cap S_2$$







## Topic: Sets



### 3. Disjoint sets

$$S_1 \cup S_2 = \{1, 2, 3, 4, 5, 6\}$$

$$S_1 \cap S_2 = \emptyset$$





## Topic: Sets



### 3. Set Difference: $S_1 - S_2$

eg:

~~Equation:~~

$$S_1: \{2, 3, 4\}$$

$$S_2: \{3, 5, 7\}$$

Element in  $S_1$  that are not in  $S_2$ .

$$\begin{aligned} S_1 - S_2 &= \{2, 3, 4\} - \{3\} \\ &= \{2, 4\} \end{aligned}$$



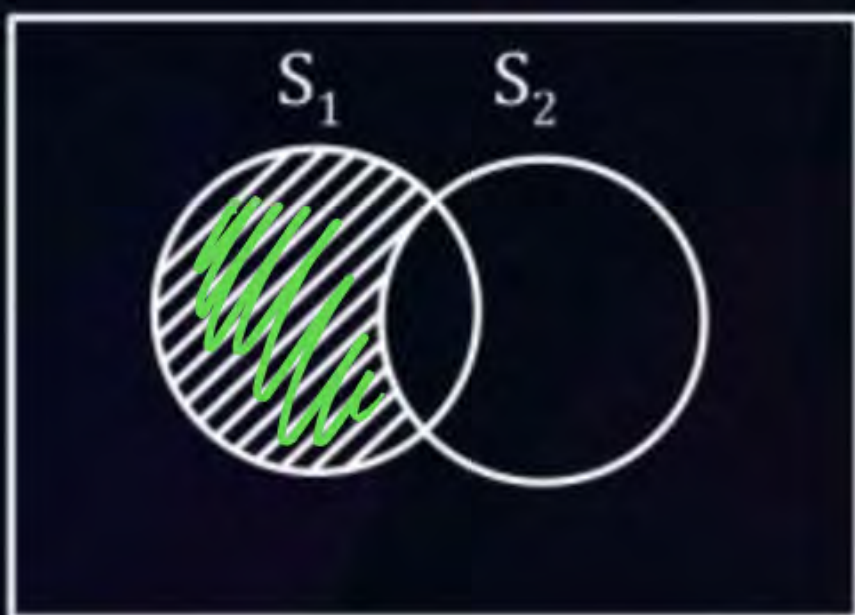


## Topic: Sets

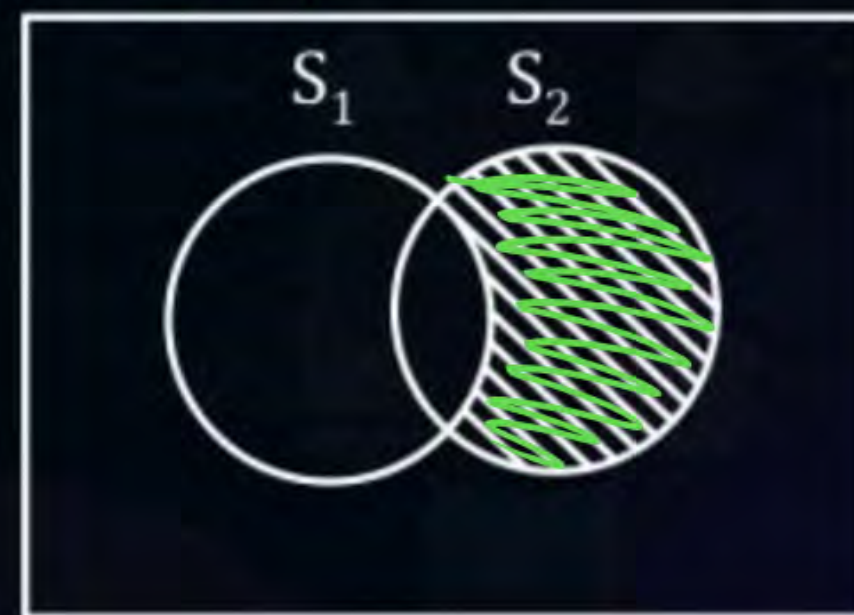
Is  $S_1 - S_2 = S_2 - S_1$

Not necessary

$S_1 - S_2$



$S_2 - S_1$   ~~$S_1 - S_1$~~





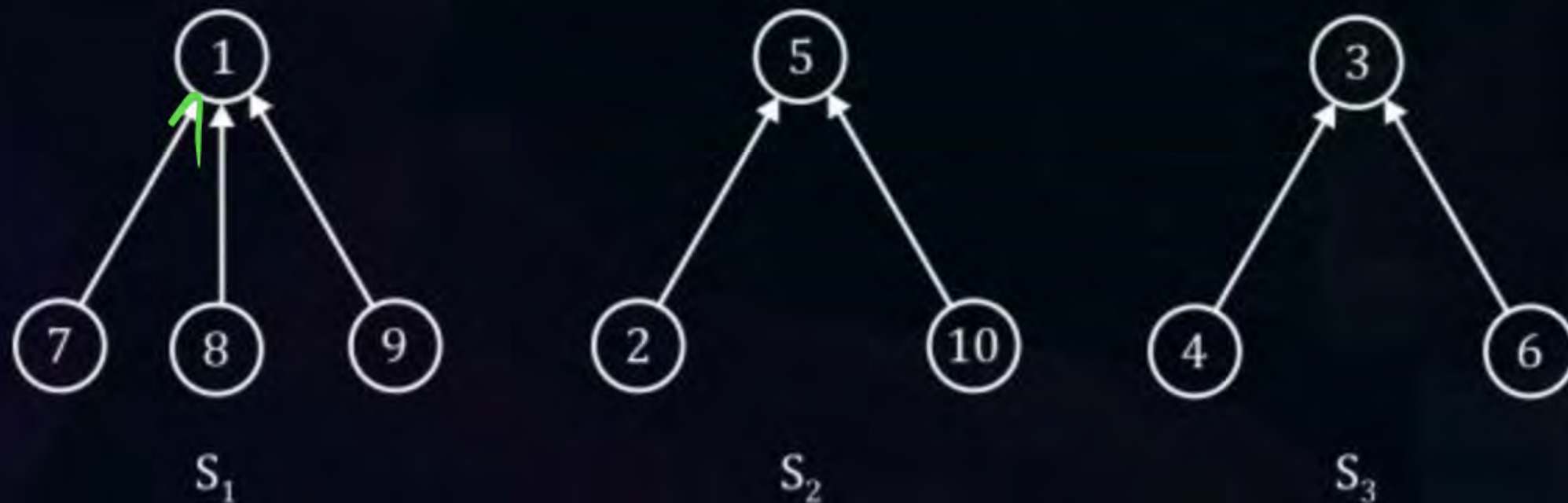


## Topic: Sets



It is assumed that the elements of the sets are the numbers 1, 2, 3, n. These number might, in practice, be indices into a symbol table in which the names of the elements are stored. We assume that the sets being represented are pairwise disjoint (that is, if  $S_i$  and  $S_j$ ,  $i \neq j$ , are two sets, then there is no element that is in both  $S_i$  and  $S_j$ ).

For example, when  $n = 10$  the elements can be partitioned into three disjoint sets,  $S_1 = \{1, 7, 8, 9\}$ ,  $S_2 = \{2, 5, 10\}$ , and  $S_3 = \{3, 4, 6\}$ . representation for these







## Topic: Sets



Tree Diagram Representation of a set:

$S_1: \{1, 7, 8, 9\}$

$S_2: \{2, 5, 10\}$

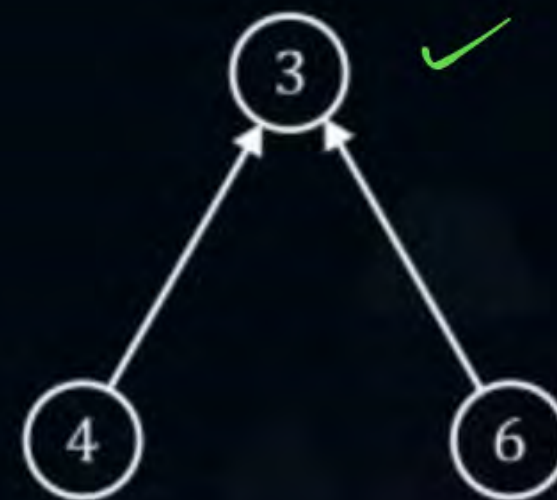
$S_3: \{3, 4, 6\}$



$S_1$



$S_2$



$S_3$

Find (8) = 1

Find (7) = 1

Find (1) = 1

Find (10) = 5

Find (3) = 3





## Topic: Sets



### Important points:

1. In tree diagram representation of a set, the Root of the tree is considered as the representative of the set.
2. Usually (not mandatory), that the 1st element of the set is the representative of that set.



## Topic: Sets



The operations we wish to perform on these sets are:

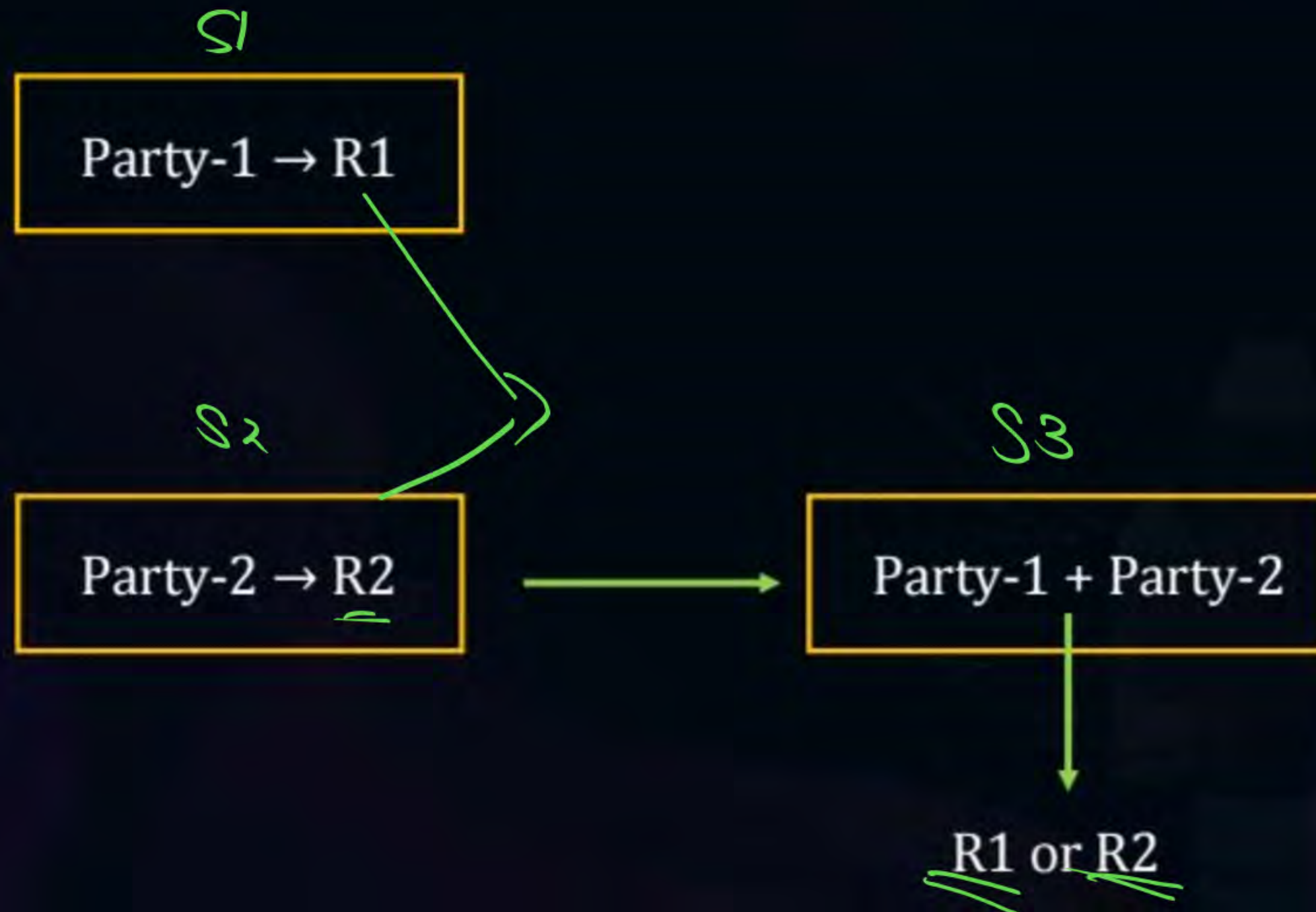
1. Disjoint set union. If  $S_i$  and  $S_j$  are two disjoint sets, then their union  $S_i \cup S_j =$  all element  $x$  such that  $x$  is in  $S_i$  or  $S_j$ . Thus  $S_1 \cup S_2 = \{1, 7, 8, 9, 2, 5, 10\}$ . Since we have assumed that all sets are disjoint, we can assume that following the union of  $S_i$  and  $S_j$  the sets  $S_i$  and  $S_j$  do not exist independently; that is, they are replaced by  $S_i \cup S_j$  in the collection of sets.
2. Find (i). Given the element  $i$ , find the set containing  $i$ . Thus, 4 is in set  $S_3$ , and 9 in set  $S_1$ .

↓  
Representative





## Topic: Sets





## Topic: Sets



$$S_1: \{1, 7, 8, 9\}$$

$$S_2: \{5, 2, 10\}$$



OR



Possible representation of  $S_1 \cup S_2$





## Topic: Sets

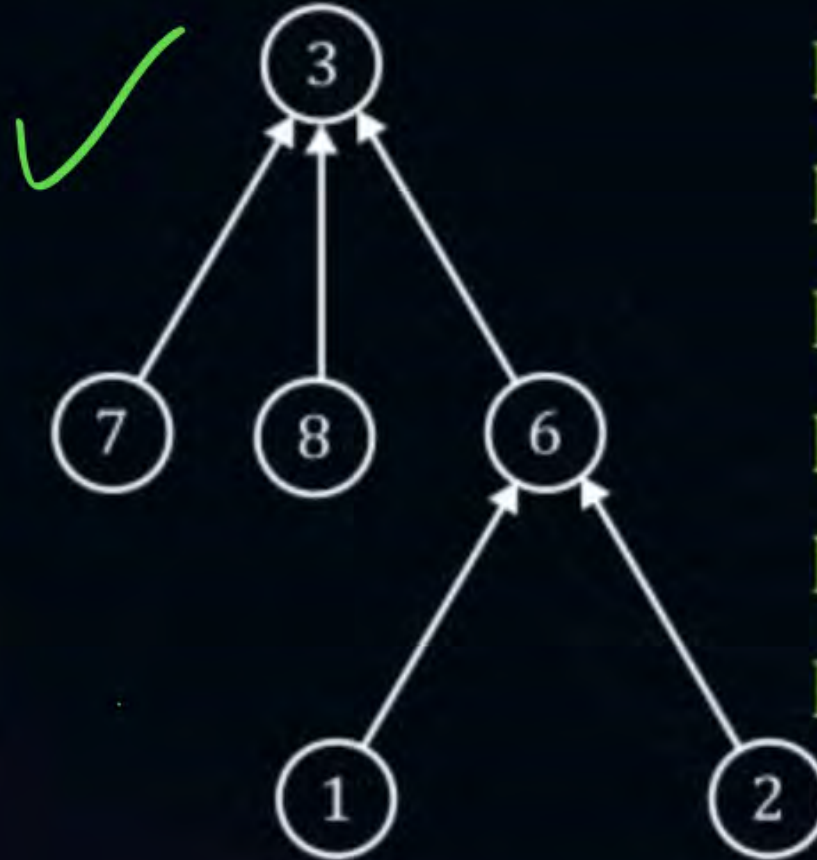
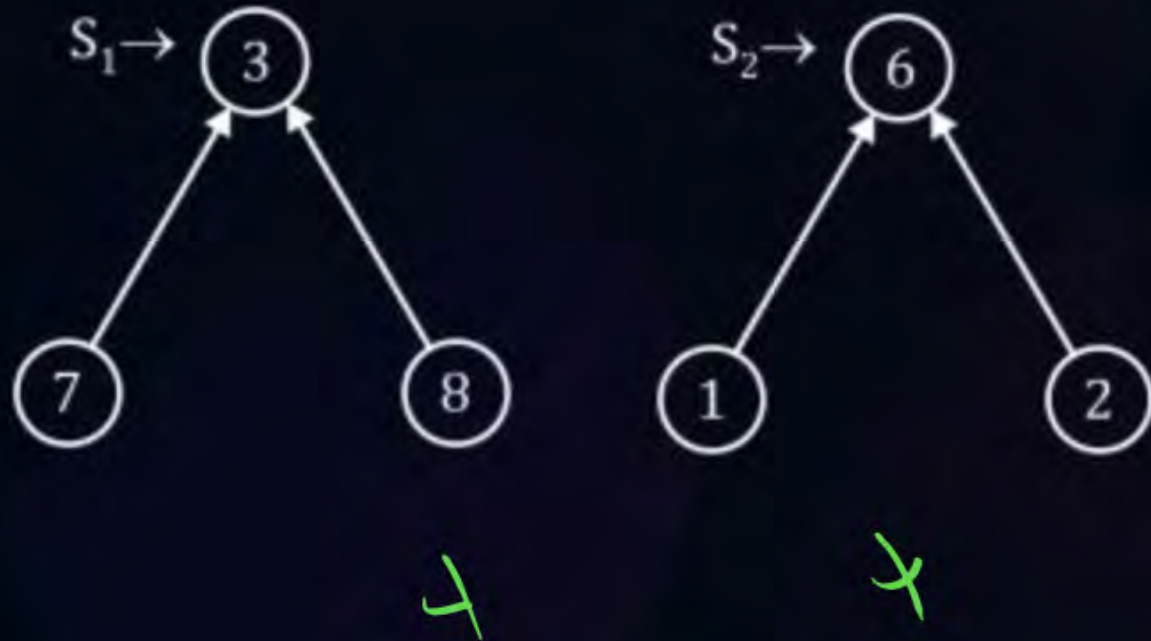


### Important:

Difference between parent and find/ Representative

$S_1: \{3, 7, 8\}$

$S_2: \{6, 1, 2\}$



Representative

Find (8) = 3

Find (6) = 3

Find (2) = 3

Find (i) always gives the root

Representation



## Topic: Sets



*Note*  
Important Role:

In above  $S_1 \cup S_2$

6 is the parent of 2 but 3 is the overall set representative.

Parent (2) = 6

Find (2) = 3

Parent (Root) = -1

Parent (3) = -1

Find (3) : 3





## Topic: Sets



Parent (Representative) = -1

Parent (Root) = -1

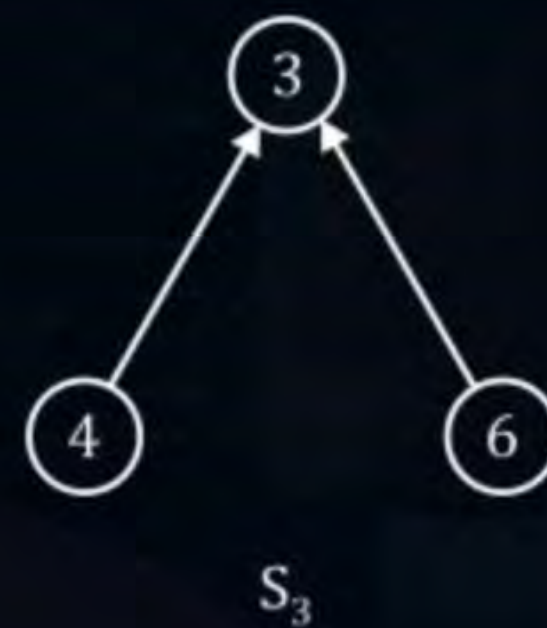
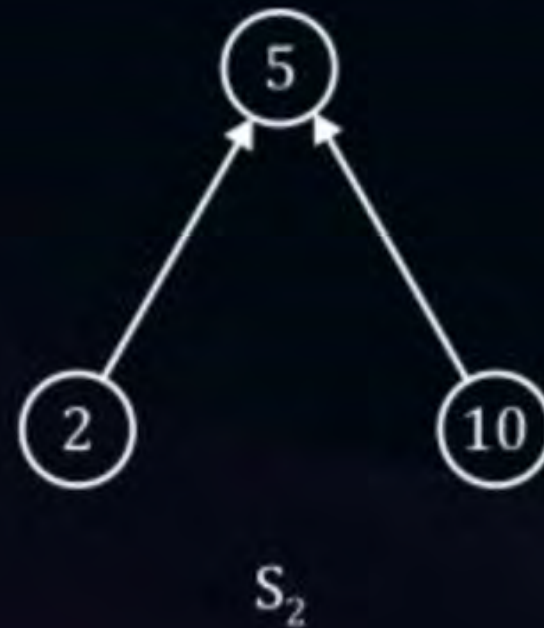
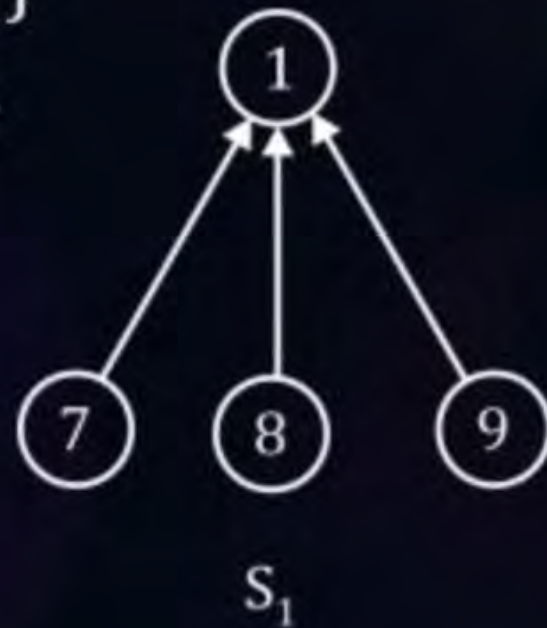
| i | [1] | [2] | [3] | [4] | [5] | [6] | [7] | [8] | [9] | [10] |
|---|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|
| P | -1  | 5   | -1  | 3   | -1  | 3   | 1   | 1   | 1   | 5    |

Array representation of  $S_1$ ,  $S_2$  and  $S_3$  of figure

$S_1$ : {1, 7, 8, 9}

$S_2$ : {2, 5, 10}

$S_3$ : {3, 4, 6}



[NAT]



#Q. Write down the array representation of the given tree diagram of set.

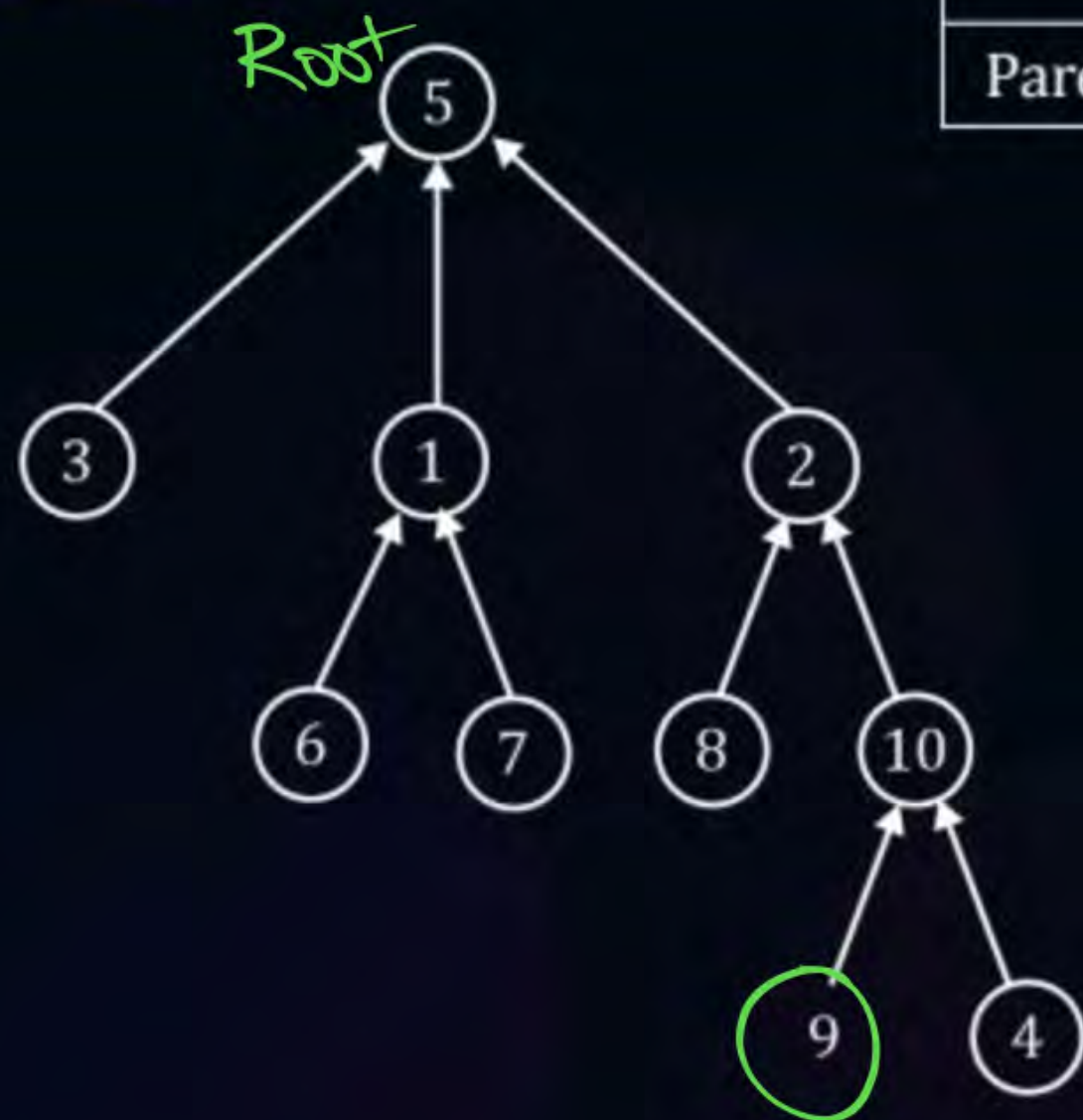




## Topic: Sets



Solution:



| i          | 1 | 2 | 3 | 4  | 5  | 6 | 7 | 8 | 9  | 10 |
|------------|---|---|---|----|----|---|---|---|----|----|
| Parent [i] | 5 | 5 | 5 | 10 | -1 | 1 | 1 | 2 | 10 | 2  |

a ← Find (4) = 5      c ← ~~Parent~~ (5) = 5  
b ← Parent (4) = 10      d ← Find (5) = 5

$$\underline{a + b + c + d = ?}$$



## Topic: Sets



1. Algorithm Union (i, j)
2. {
3.      $P[i] = j;$
4. }

$S_1 \rightarrow i$   
 $S_2 \rightarrow j$

- 
1. Algorithm find (i)
  2. {
  3.     While ( $p[i] \geq 0$ ) do  $i = p[i];$
  4.     return  $i$
  5. }



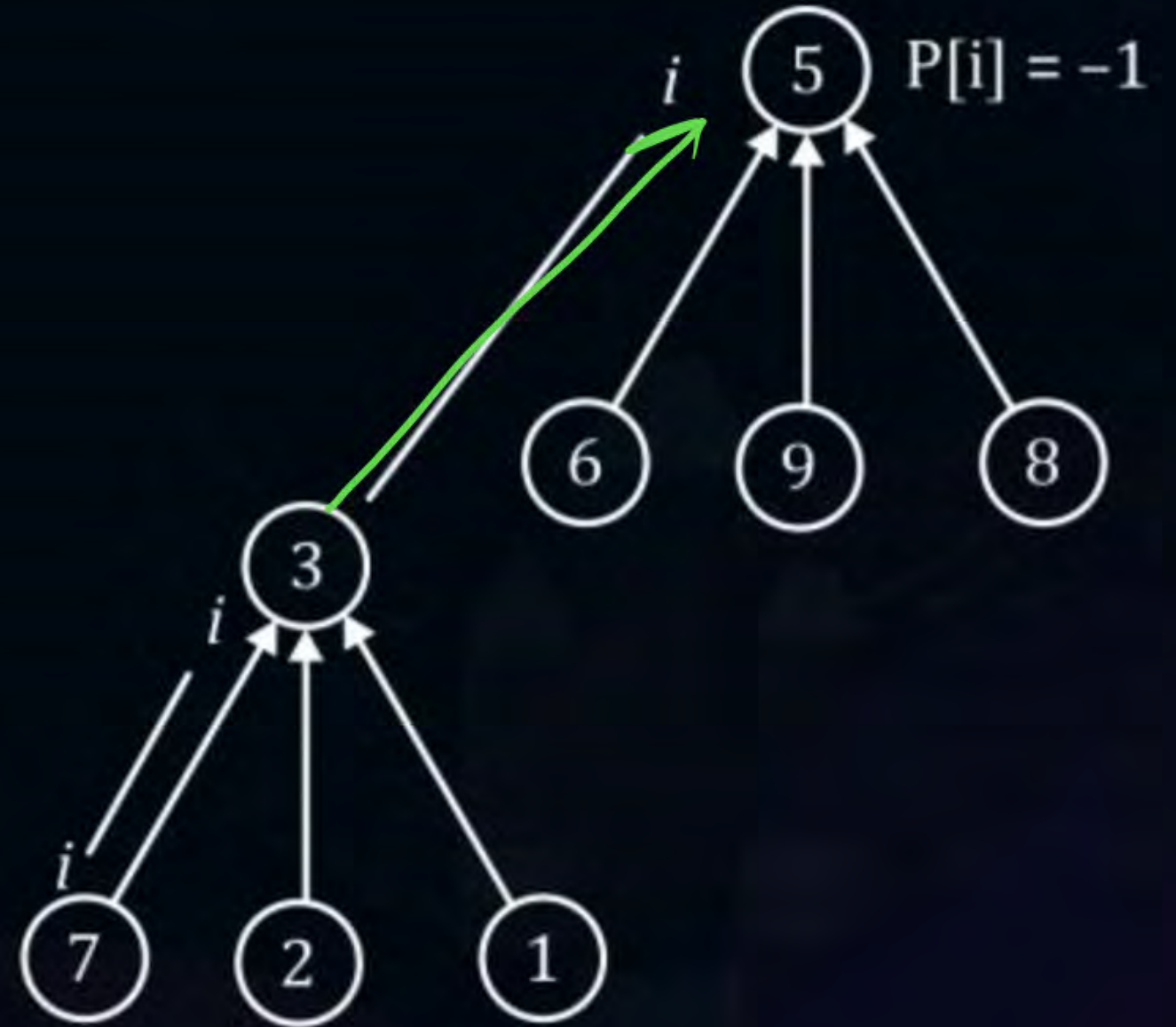




## Topic: Sets



union





## Topic: Sets



$i = 7$

Find (7)

$i = 7 \ P[i] = 3 \geq 0$

$i = 3 \ P[3] = 5 \geq 0$

$i = 5 \ P[5] = -1 \geq 0$  (False)

exit loop

return (5)







## Topic: Greedy Method



1. Algorithm Kruskal [E, cost n, t] → temp array
2. {
3. Construct a min Heap out of the edge cost using Heapify
4. For i = 1 to n do parent [i] := -1;
5. i : 0; mincost: = 0.0;
6. While ((i < n - 1) and (heap not empty)) do
7. {
- 8. Delete a minimum cost edge (u,v) from the heap and reheapify using adjust;
9. J = find (u); k = find (v);
10. If (j ≠ k then)
11. { → avoid Cycle
12. i: i + 1 ✓



## Topic: Greedy Method



```
13.   T[i,1]:=u; [i1, 2 ]=v
14.   Mincost := mincost + cost [u, v]
15.       Union (j ,k)
→16.   }
17.   }
18.       If (i ≠ n-1) then write ("No spanning tree")
19.       Else return mincost;
20.   }
```





## Topic: Sets



Code walk through:

Given graph: edge lost: [ ~~9~~, ~~11~~, ~~13~~, ~~15~~, ~~17~~, ~~21~~, ~~23~~, ~~24~~, ~~27~~ ]

Min heap of edge cots of given graph.



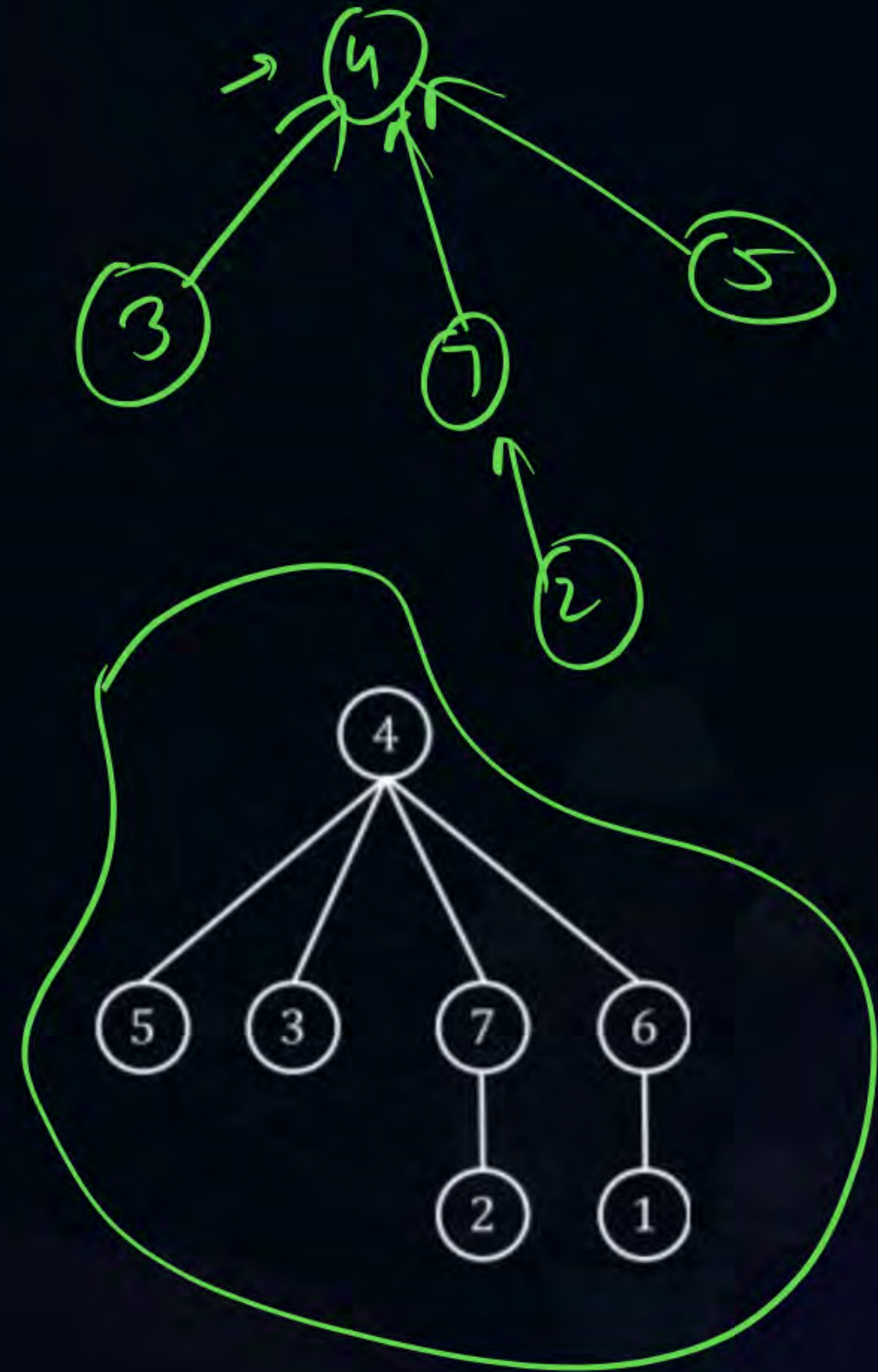
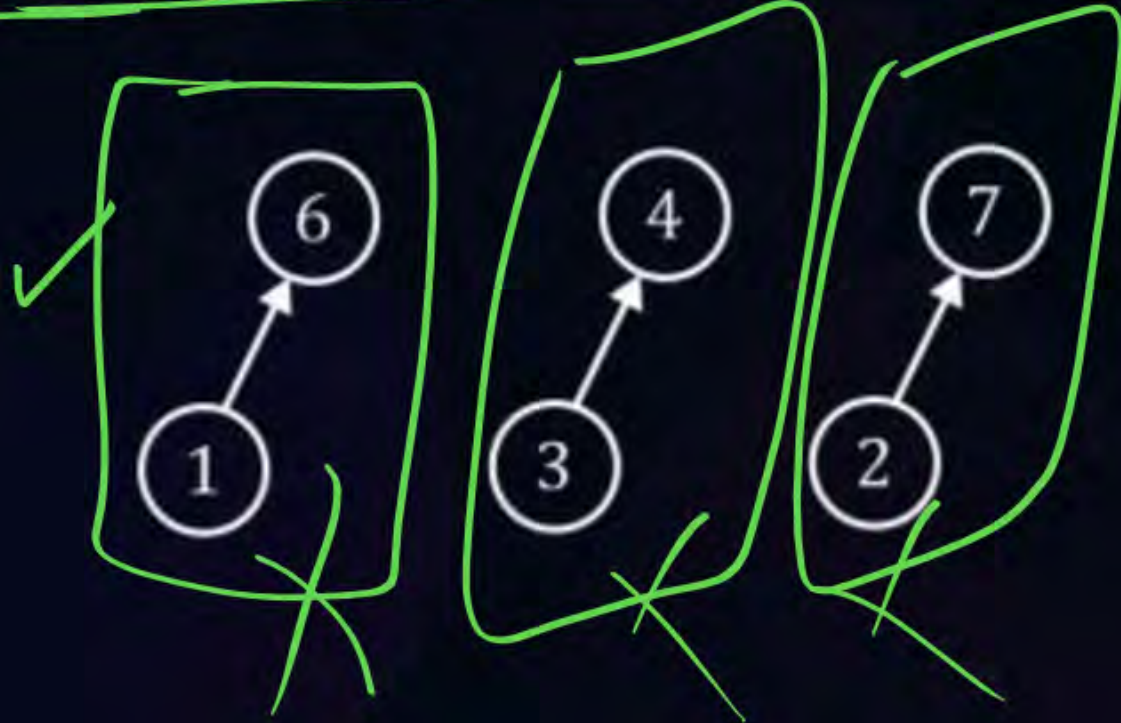
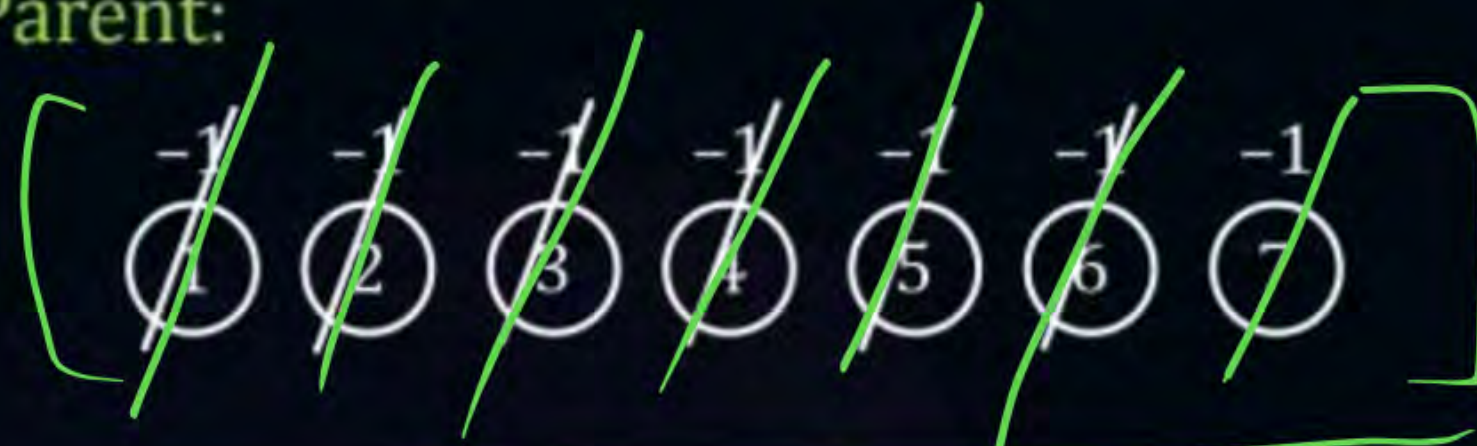




## Topic: Sets

Initially

Parent:



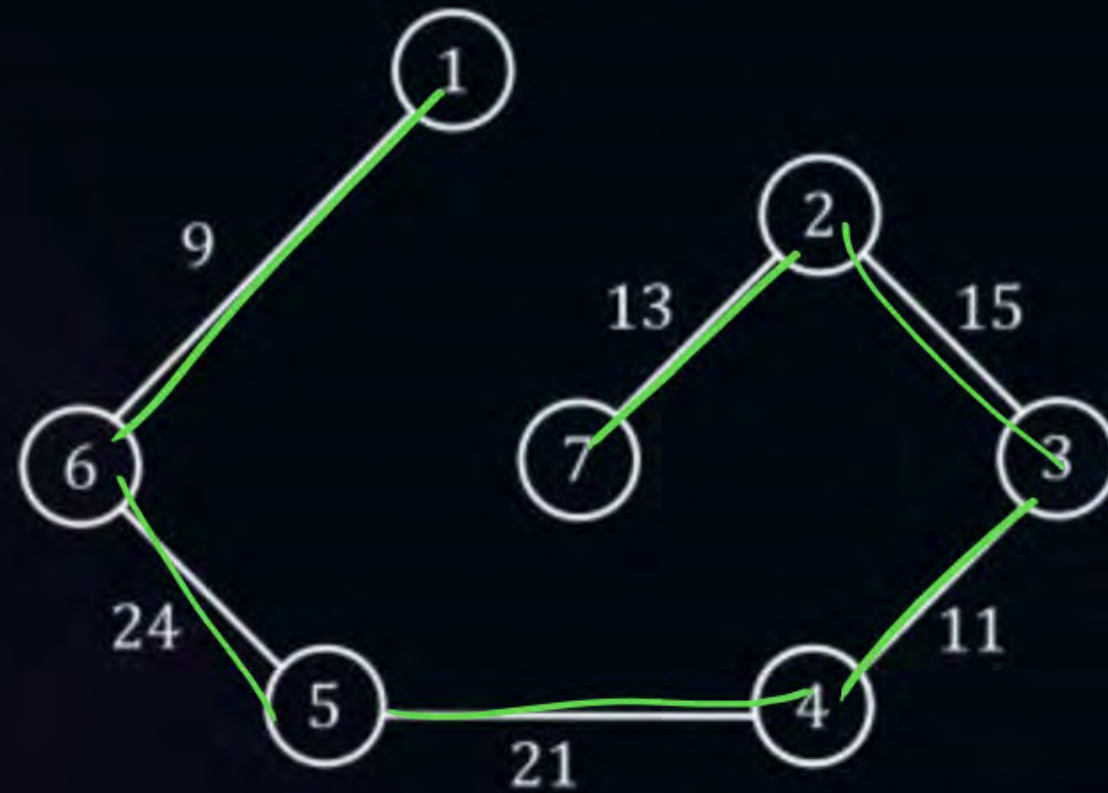




## Topic: Sets



Min cost: =  $0 + 9 + 11 + 13 + 15 + 21 + 24 = 93$





## Topic: Sets



$$1. \quad (u, v) = (\underbrace{1}_u, \underbrace{6}_v)$$

$$j = \text{find}(1) = \underline{1}$$

$$k = \text{find}(6) = \underline{6}$$

$$\begin{array}{l} \underline{j \neq k} \\ \left. \begin{array}{l} t[\underline{1}, 1] = 1 \\ t[\underline{1}, 2] = 6 \end{array} \right\} \\ \text{Union } (1, 6) \end{array}$$

1st

$$2. \quad (u, v) = (\underbrace{3}_u, \underbrace{4}_v)$$

$$j = \text{find}(3) = 3$$

$$k = \text{find}(4) = 4$$

$$\begin{array}{l} j \neq k \\ t[\underline{2}, 1] = 3 \\ t[\underline{2}, 2] = 4 \\ \underline{\text{Union } (3, 4)} \end{array}$$





## Topic: Sets



3.  $(u, v) = (2, 7)$

$$j = \text{find}(2) = 2$$

$$k = \text{find}(7) = 7$$

$$j \neq k$$

$$t[3, 1] = 2$$

$$t[3, 2] = 7$$

$$\text{Union}(2, 7)$$

4.  $(u, v) = (2, 3)$

$$j = \text{find}(2) = 7$$

$$k = \text{find}(3) = 4$$

$$j \neq k$$

$$t[4, 1] = 2$$

$$t[4, 2] = 3$$

$$\text{Union}(7, 4)$$



## Topic: Sets



5.  $(u, v) = (7, 4)$

$$j = \text{find}(7) = 4$$

$$k = \text{find}(4) = 4$$

$$\frac{j = k}{(\text{cycle})}$$

6.  $(u, v) = (5, 4)$

$$j = \text{find}(5) = 5$$

$$k = \text{find}(4) = 4$$

$$j \neq k$$

$$t[5, 1] = 5 \quad \checkmark$$

$$t[5, 2] = 4 \quad \checkmark$$

$$\underline{\text{Union}(5, 4)}$$







## Topic: Sets



$$7. (u, v) = (7, 5)$$

$$j = \text{find}(7) = 4$$

$$k = \text{find}(5) = 4$$

$$j = k$$

(cycle)

$$8. (u, v) = (5, 6)$$

$$j = \text{find}(6) = 6$$

$$k = \text{find}(5) = 4$$

$$j \neq k$$

$$t[6, 1] = 6$$

$$t[6, 2] = 5$$

$$\text{Union}(6, 4)$$



## Topic: Sets



Time and space complexity analysis of Kruskal (overall).

$e$  = number of edges in given graph

1. Time complexity =  $O(e \log e)$
2. Space complexity =  $O(e + v)$   
 $= O(e + n)$





## Topic: Sets



Kruskal logic:

Given a graph:



$$n = |V| = 7$$

$$\begin{aligned}\text{MCST edges} &= n - 1 \\ &= 6\end{aligned}$$

Cost of MCST

$$= 9 + 11 + 13 + 15 + 21 + 24 = 93$$



TIPS :-

- 1) Revision
- 2) Practice
  - ↳ DPP
  - ↳ WT
  - ↳ PYQs

Test Series





## 2 mins Summary



Miscellaneous

Set ✓



**THANK - YOU**

